PHYSICAL AI INSPECTION PLATFORM · 01 CAPTURE

IntCloud Optic

An AI optical inspection module that captures defects so they can be seen

It searches a wide area, then re-aims at higher magnification to look closely at anything suspicious
Registered patent: wide-angle image synthesis for rotating objects · applied in display, film and paper



IntCloud Optic optical module (design visualization) and a concept diagram: search wide, then look again closer

02 — OVERVIEW

Inspection starts with a camera that captures defects well

IntCloud Optic is the Capture step of the INTELLIZ physical AI inspection platform. It first shoots a wide area to find suspicious spots, then raises the magnification and aims again to look closely.

FIG. 01 MODULE · INTCLOUD OPTIC



IntCloud Optic optical inspection module · design visualization



Imaging a cup · 3D rendering example

01 Capture

Step one of the inspection platform

Capture, AI judgment, pre-install validation, operation

1 registered patent

Wide-angle image synthesis

2 filed: multi-magnification aiming

3 sites

Display · Film · Paper

Paper rewinder: trial (PoC) stage

IN · WHAT GOES IN

The product to inspect

Products passing on the line are shot wide in one go; suspicious spots are shot again at higher magnification.

OUT · WHAT COMES OUT

Images that capture the defects

IntLearn Pro judges the images with AI; IntScan Pro and IntViewer run the inspection and show the results.

A defect you cannot see, you cannot find.

So it searches wide, then looks again at higher magnification at anything suspicious.

03 — WIDE · CLOSE

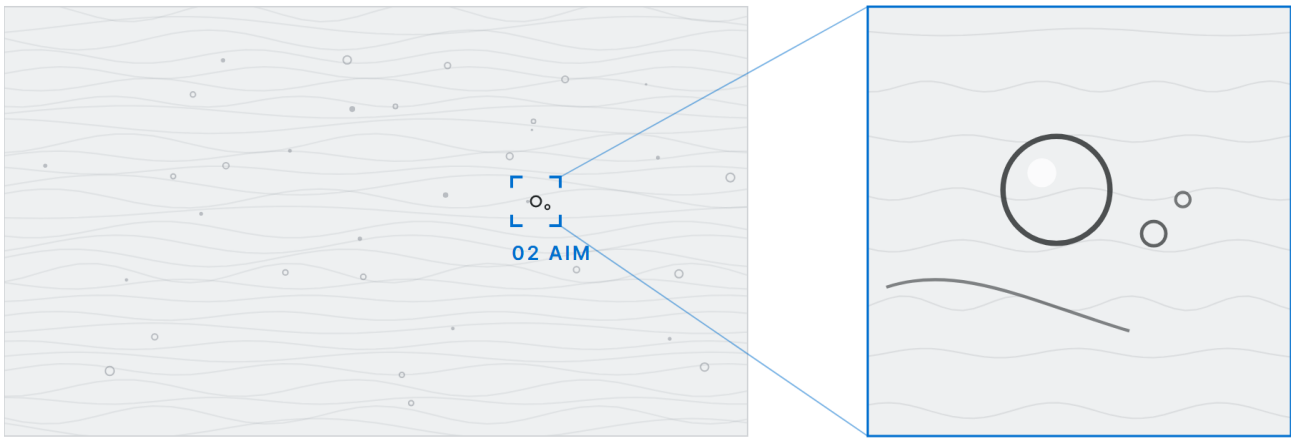
Search wide, then re-aim at suspicious spots

A wide view (wide angle, low magnification) first shoots a wide area in one go. Where a spot looks suspicious, the module raises the magnification, aims again and confirms even fine defects.

FIG. 02 WIDE → AIM → CLOSE · CONCEPT

01 WIDE **Search wide**
Wide angle, low magnification: a wide area in one shot

03 CLOSE **Look closely**
Re-aim closer to see fine defects



A wide area at once · defects look small

A narrow area · defects look large

Concept diagram (example): find a suspect spot in the wide view, then re-aim at higher magnification

Key features

Product page 01 FEATURES

01	Search wide	Captures a wide area in one shot with a wide-angle, low-magnification view.
02	Look closely	Re-aims at suspicious spots with high magnification to confirm even fine defects.

WIDE · SHOOTING WIDE

You see a lot at once

A wide area in one image, but defects look small.

CLOSE · SHOOTING CLOSE

Small defects look large

Defects look large, but each image covers less.

Search wide, and go close only where needed.

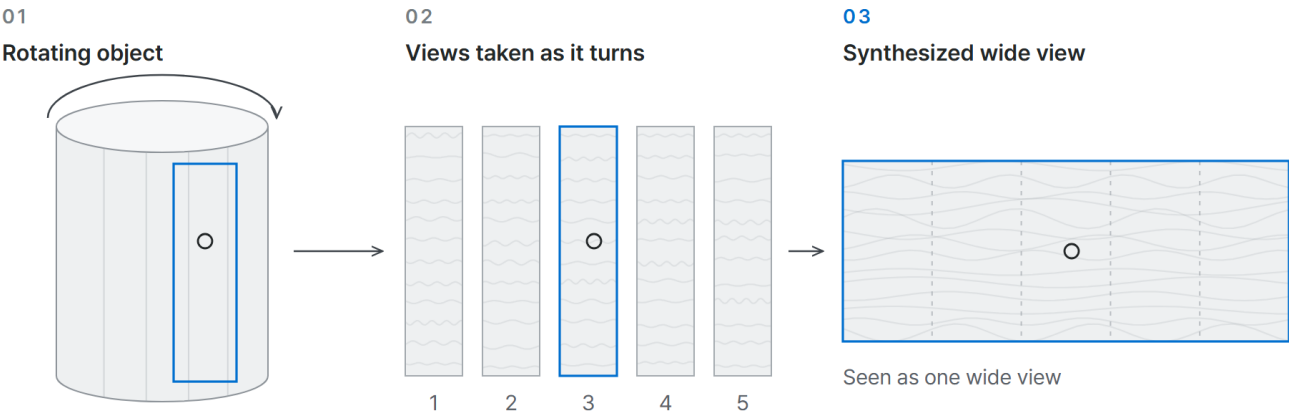
Wide shots and close shots are kept apart, so the wide area and the small defects are both covered.

04 — PATENTS

Rotating parts are seen in one synthesized wide view

Wide-angle image synthesis for rotating objects (one wide view of a turning part) is a registered patent. Multi-magnification aiming (aiming at different magnifications) has two applications filed.

FIG. 03 WIDE-ANGLE SYNTHESIS · CONCEPT



Concept diagram: a rotating object synthesized into one wide view · the actual processing may differ

Patents Product page 01 FEATURES

Wide-angle image synthesis for rotating objects	Registered · No. 10-2521516
Multi-magnification aiming	Filed · Application No. 10-2025-0157093
Multi-magnification aiming	Filed · Application No. 10-2025-0157187

FIG. 04 FOUR SIDES · DESIGN VISUALIZATION



IntCloud Optic seen from four sides · design visualization

Even a turning part, in one wide view.

Wide-angle image synthesis is a registered patent; multi-magnification aiming has two applications filed.

05 — PLATFORM · CONNECTED PRODUCTS

IntLearn Pro judges the captured images

IntLearn Pro judges the images with AI; IntScan Pro and IntViewer run inspections and show results. Before installing, check imaging conditions and performance in an IntLearn Kinesis virtual cell.

FIG. 05 CAPTURE → JUDGE → RUN → VIEW

Pre-install validation **IntLearn Kinesis**
Imaging conditions and inspection performance can be checked in a virtual cell

01 Capture	02 Train	03 Package	04 Line inspection	05 Review results
IntCloud Optic	**IntLearn Pro**	**.izpkg**	**IntScan Pro IntSight**	**IntViewer Pro**
Images that capture the defects	Trains on the images, judges with AI	Model and criteria in one file	Inspect on the line	Results shown on screen

Retrain IntLearn Pro on line data

Where IntCloud Optic sits, from consultation to adoption at INTELLIZ

FIG. 06 INTLEARN PRO · KINESIS · INTWISE



02
IntLearn Pro
Deep learning visual inspection software



03 New in 2026
IntLearn Kinesis
Validate in a virtual factory before you install



04 New in 2026
IntWise
Operate by speaking, and the know-how becomes a company asset

For film foreign matter and bubbles

Guide · Look wide, then close in on anything suspect

INTLEARN PRO · DETECTION ENGINE

Where, and how many

IntLearn Pro's detection engine (IntDetection) puts a box on each bubble and piece of foreign matter.

INTSCAN PRO · INTVIEWER PRO

Results on record and on screen

Results, logs and images stay together in IntScan Pro; results in operation are viewed in IntViewer Pro.

Capture, AI judgment, pre-install validation, operation.

Of the four products on the INTELLIZ physical AI inspection platform, IntCloud Optic does the Capture step.

06 — EVIDENCE · PROOF POINTS

Optics and imaging patents, and display, film and paper sites

Optics and imaging patents: 2 registered, 4 filed. Sites are display, film and paper: film foreign matter and bubble inspection, and paper rewinder (roll rewinding) inspection at the trial (PoC) stage.

2 registered · 4 filed

Optics and imaging patents

Display · Film · Paper

Applied sites

Film foreign matter and bubble inspection, paper rewinder inspection (PoC)

INTELLIZ track record · film foreign matter and bubble inspection rated suitable in an external evaluation for intelligent processes in Korea's root industries

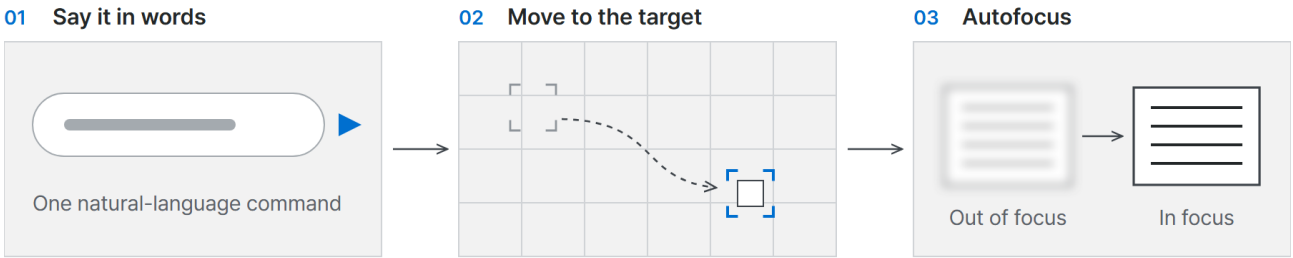
1. The image is a design visualization. 2. Paper rewinder inspection is at the PoC stage.

Product video

Product page 04 VIDEO · 30-second demo

Move to a target and bring it into focus with a natural-language command, shown on real equipment.

FIG. 07 VIDEO · 30 S DEMO



Concept diagram of the product video "Natural Language Control & Autofocus" · not frames from the video

Product video · QR · 30 s

Natural Language Control & Autofocus

Intelliz 30-Second Demo · YouTube

youtu.be/CWPHQbXex8o

IntWise · QR

Operate by speaking

IntWise page: videos show camera exposure, lighting and focus adjusted in natural language.

intelliz.net/en/products/intwise

One sentence to move; focus comes by itself.

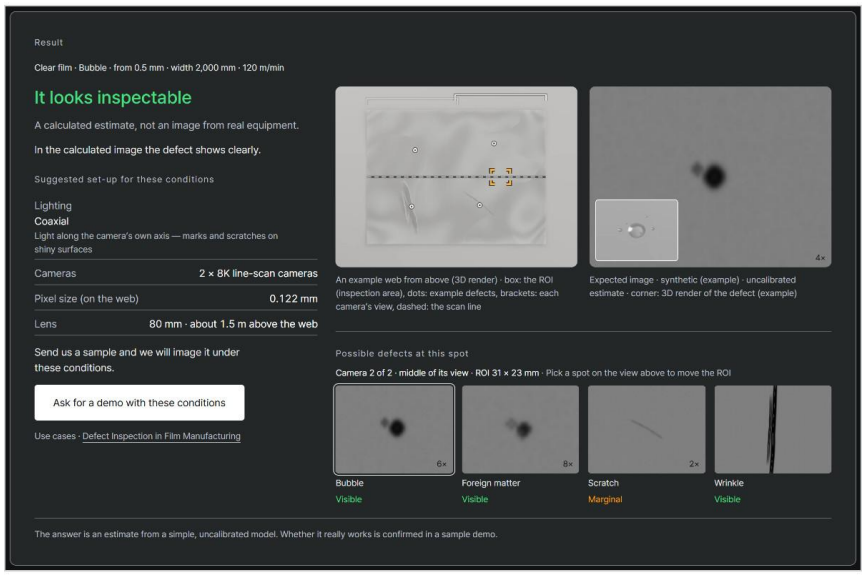
IntCloud Optic product video · Intelliz optical control demo on real equipment (30 s)

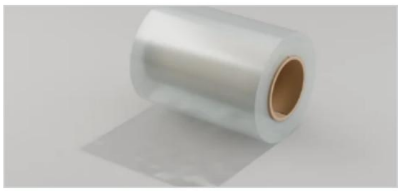
07 — NEXT STEPS · OPTICS LAB

Before you shoot, check first what will show

In the IntScan Pro Optics Lab, pick clear film and bubble: a calculated estimate shows at once whether line scan (imaging moving film one line at a time) can see it, with a suggested set-up.

FIG. 08 OPTICS LAB · CLEAR FILM · BUBBLE





CONDITIONS PICKED

Clear film · Bubble

from 0.5 mm · width 2,000 mm
120 m/min

"It looks inspectable"

A calculated estimate, not an image from real equipment.

Uncalibrated estimate · a sample demo confirms whether it really works.

IntScan Pro Optics Lab result (intelliz.net) · the suggested set-up is a calculated example, not a spec

Which defects show depends on the light

Guide · film bubble and scratch inspection

TRANSMITTED LIGHT

Light from behind

Bubbles and foreign matter usually show as shadows.

ANGLED LIGHT

Light from an angle

Scratches on the surface usually show up well.

COAXIAL LIGHT

Along the camera axis

Dents and scratches on glossy film are sometimes caught well.

Optics Lab · QR

Clear film · Bubble

Line-scan feasibility and sample demo

intelliz.net/en/solutions/optics-lab

Guide · QR

Film Bubble and Scratch Inspection: What to Prepare

intelliz.net/en/guides

An imaging review starts with four numbers.

Film width · line speed · smallest defect to find · how often and where defects occur (guide)



IntCloud Optic · search wide, look closer · design visualization, concept diagram

08 — CONTACT · NEXT STEPS

Considering a similar inspection?

Including the following in your inquiry helps us prepare our answer.

01	What you inspect: material, shape and size	02	The defects to find and your pass/fail criteria
03	Line speed and how products are conveyed	04	Space available for cameras and lighting
05	Where results need to go (PLC, reject unit, etc.)		

Next steps

Where the product page leads · intelliz.net

Optics Lab	Clear film · Bubble — see first whether line scan can inspect it	intelliz.net/en/solutions/optics-lab
Inspection guides	Film Bubble and Scratch Inspection: What to Prepare	intelliz.net/en/guides/film-bubble-scratch-imaging-prep
Getting ready	AI Inspection Adoption Checklist	intelliz.net/en/tools/inspection-checklist
FAQ	What products do you offer?	intelliz.net/en/company/faq
Solution Experience	Try IntCloud Optic · an example view of a virtual factory (digital twin)	intelliz.net/en/solutions/experience

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Manufacturing Lab B-403, Hyundai Silicon Alley Dongtan, 150, Dongtanyeongcheon-ro, Hwaseong-si, Gyeonggi-do
Japan distributor BIT DEVICE Co., Ltd., 6-3-24-L430 Nishinakajima, Yodogawa-ku, Osaka

Product page

Optics Lab

INTELLIZ

© INTELLIZ Corp. 2026 · Product information follows public material at intelliz.net (2026-11-03).
IntCloud Optic images are design visualizations; the inspection scene is a 3D rendering example.
Cover, back cover and FIG. 02, 03, 05, 07 are concept diagrams, not images or values from the module.
The Optics Lab screen is an uncalibrated estimate; a sample demo confirms what really works.
Paper rewinder inspection is at the PoC stage.